

Industrial LoRaWAN DTU for robust serial data connectivity



LT312 Series LoRaWAN Terminal Node

• LoRaWAN 1.0.3

• RS-232/RS-485

• Modbus-RTU

• Industrial DTU

1. Product Overview

LT312 is a versatile industrial LoRaWAN DTU for serial data acquisition and transparent transmission across building, campus, and factory scenarios.

Key Features:

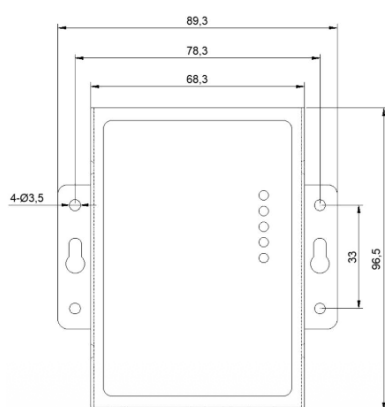
- **Industrial serial robustness:** Isolated RS-232/RS-485 with built-in 15 kV ESD protection
- **Flexible LoRaWAN operation:** LoRaWAN 1.0.3, Class A/C, OTAA/ABP, ADR or fixed SF
- **Strong RF coverage:** Up to 22 dBm Tx power and -135 dBm sensitivity at SF12/BW125 kHz
- **Multi-region deployment:** EU433/CN470/CN779/EU868/AS923/AU915/KR920 coverage
- **Task orchestration:** Script editing, heartbeat customization, polling and watchdog support

Core Technical Specifications

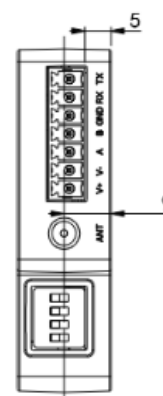
Technical Indicator	Specification
Device Operating Modes	Class A/C, supports OTAA and ABP
Adaptive	Supports ADR and fixed SF switching
Serial Data Services	Transparent transmission
Automation	Supports script editing
Cloud Integration Capability	End-to-cloud data forwarding through LoRaWAN gateways/platforms
Frequency Bands	EU433/CN470/CN779/EU868/AS923/AU915/KR920

Technical Indicator	Specification
Transmit Power	Up to 22 dBm transmit power, -135 dBm sensitivity @ SF12/BW125kHz
Serial Interfaces	1 x RS-232 (DB9) + 1 x RS-485 (3.81 mm 2-pin)
Power Input	DC12V/1A
Power Consumption	Average power consumption ≤ 0.5 W
Humidity	5 to 95% RH (non-condensing)

2. Product Dimensions



Device View



Port Side View

Notes:

1. All dimensions are in millimeters (mm), with inches (in) in parentheses.
2. All dimensions are approximate and for reference only.
3. Drawings **must not be used for manufacturing**.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category / Parameter	Specification
LoRaWAN RF	
Protocol	LoRaWAN 1.0.3
Frequency bands	EU433, CN470, CN779, EU868, AS923, AU915, KR920
Transmit power	22 dBm
Receiver sensitivity	-135 dBm @ SF12 BW125kHz
Join/operation modes	OTAA/ABP, Class A/C
Spreading factor	SF7-SF12, 6 levels adjustable
Data rate (typical)	5.5 / 3.1 / 1.8 / 1.0 / 0.6 / 0.3 kbps
Serial Interfaces	
RS-485 quantity/type	1 x 3.81 mm 2-pin terminal
RS-485 baud rate	2400-115200 bps, default 9600 bps, 8N1
RS-485 protection	Fully isolated, built-in 15 kV ESD
RS-232 quantity/type	1 x DB9
RS-232 baud rate	2400-115200 bps, default 115200 bps, 8N1
RS-232 protection	Fully isolated, built-in 15 kV ESD
Power	
Input	DC12V, 1A
Connector type	Port 1: 3.81 mm 2-pin terminal; Port 2: DC 5.5 x 2.1 mm
Power consumption	Average <= 0.5 W

Category / Parameter	Specification
Mechanical & Environment	
Storage temperature	-40 °C ~ 85 °C (-40 °F ~ 185 °F)
Operating temperature	-20 °C ~ 70 °C (-4 °F ~ 158 °F)
Ambient humidity	5 to 95% (non-condensing)
Indicators	
POWER indicator	On: normal power; Off: undervoltage or fault
RUN indicator	Blinks: normal operation; steady on/off: fault
LoRa indicator	On: LoRa data transmitting; Off: idle
RS-232 indicator	Blinks: RS-232 data received; Off: idle
RS-485 indicator	Blinks: RS-485 data received; Off: idle

4. Software Specifications

Category / Parameter	Specification
Network Features	
Protocol stack	LoRaWAN 1.0.3
Device class	Class A/C
Join method	OTAA / ABP
Link adaptation	ADR and fixed SF switching
Channel/power control	Adjustable channel and transmit power

Category / Parameter	Specification
Serial Data Services	
Data mode	Transparent transmission
Protocol compatibility	Modbus-RTU compatible polling/commands
Serial configuration	Flexible serial settings for diverse devices
Task scheduling	Scheduled tasks and active polling
Automation & Reliability	
Script support	Script editing and task orchestration
Heartbeat	Customizable LoRaWAN heartbeat
I/O output control	Configurable output behavior
Watchdog	Built-in reliability watchdog
Deployment & Integration	
Typical scenarios	Buildings, campuses, factories
Device integration	Meters, PLCs, and serial devices
Cloud path	End-to-cloud data forwarding via LoRaWAN gateway/platform

5. Ordering Information

Model Rule

Model code: LT312-<REGION>

<REGION>: Frequency band / regional variant

Product Models

Model	Region	<REGION>	Serial Port
LT312-US915	North America	US915, Tx power: 22 dBm	1 x RS-232 + 1 x RS-485
LT312-EU868	Europe	EU868, Tx power: 22 dBm	1 x RS-232 + 1 x RS-485
LT312-CN470	China	CN470, Tx power: 22 dBm	1 x RS-232 + 1 x RS-485
LT312-AS923	Asia-Pacific (except KR/IN)	AS923, Tx power: 22 dBm	1 x RS-232 + 1 x RS-485

6. Contact Us

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